Deep Learning for the Ice Properties Measurement

using Simulation Images of the IceCube Upgrade Camera System

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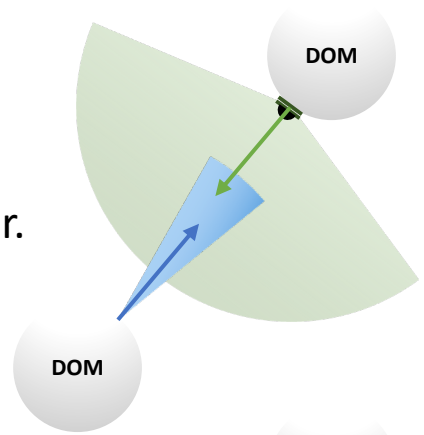
Introduction

Previously, two CNN model was developed to measure scattering length in two different geometric cases

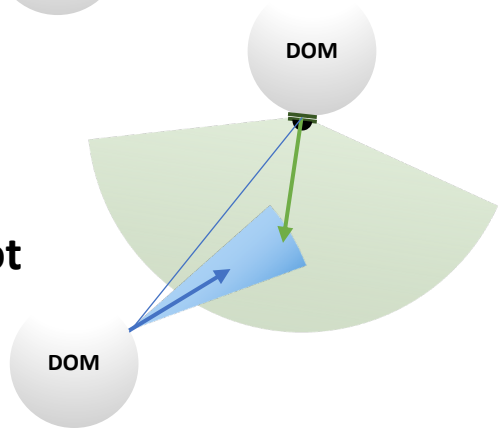
Case A

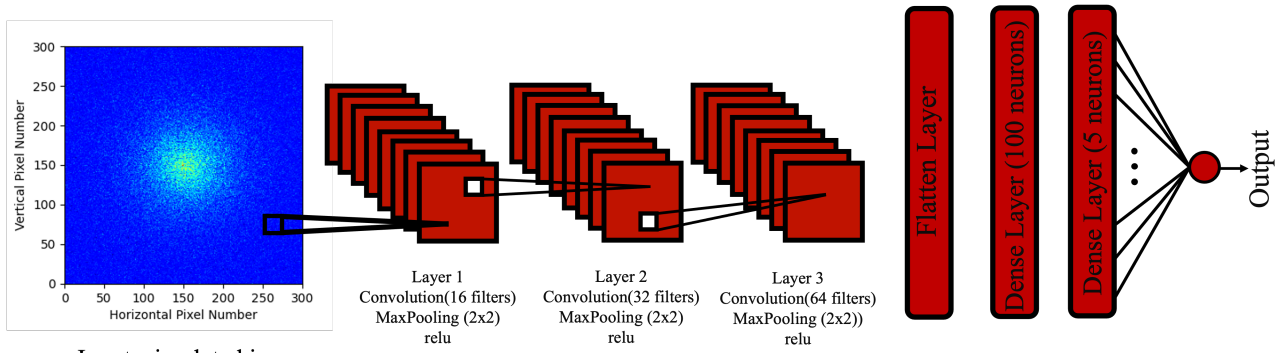
Camera and LED are **"Aligned"** to each other.



Case N

Camera and LED are **"Not aligned"** to each other.





- Input: simulated image
- Output: scattering length

Model Performance (Prediction Accuracy)

Model	Mean Error	68% of Error	95% of Error
A	0.00m	0.15m	0.30m
N	0.00m	0.22m	0.44m

240523, Calibration Call, Minje Park

This result demonstrates that using Deep Learning, we can measure ice properties with high accuracy.

As a starting point, Bulk Ice Model version 1.0 & Hole Ice Model version 1.0 has been developed

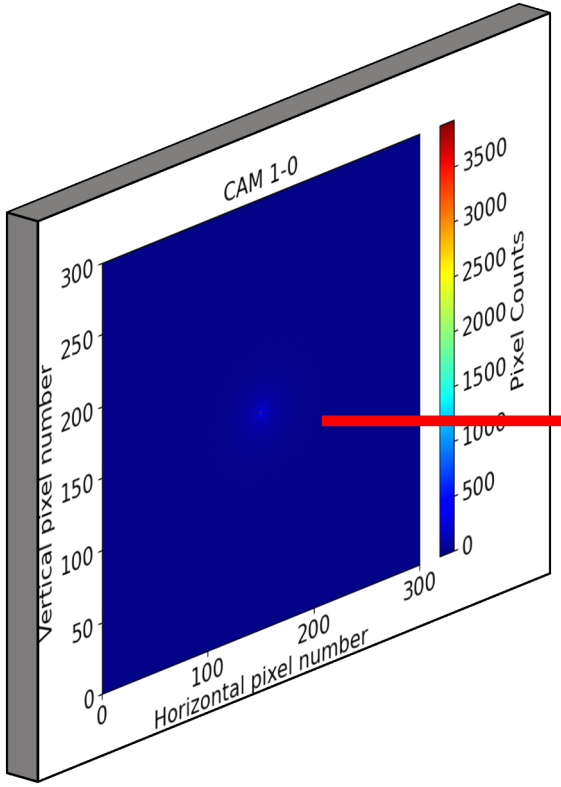
Bulk Ice Model – *version 1.0*

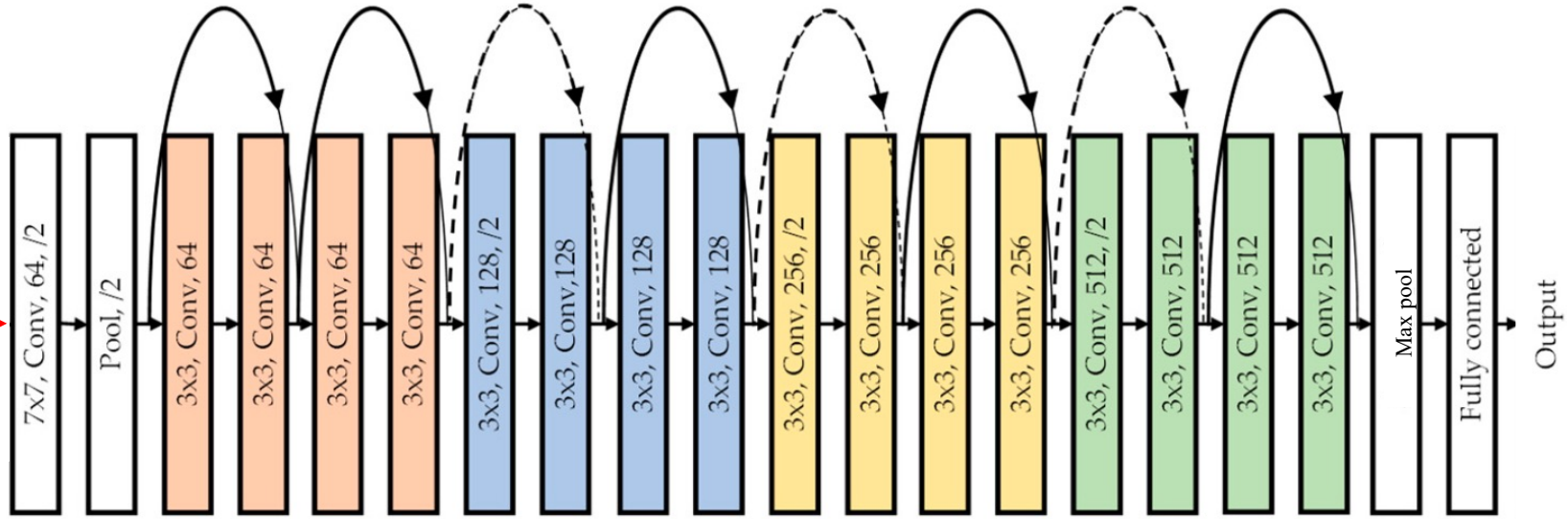
Purpose : Measure Bulk Ice Scattering Length Column Scattering from the simulation image

Architecture: Customed ResNet18

Main Constraint : (Camera LED distance : 29.8m) & (Horizontal Camera & LED)

[*distance between string 87-89*]



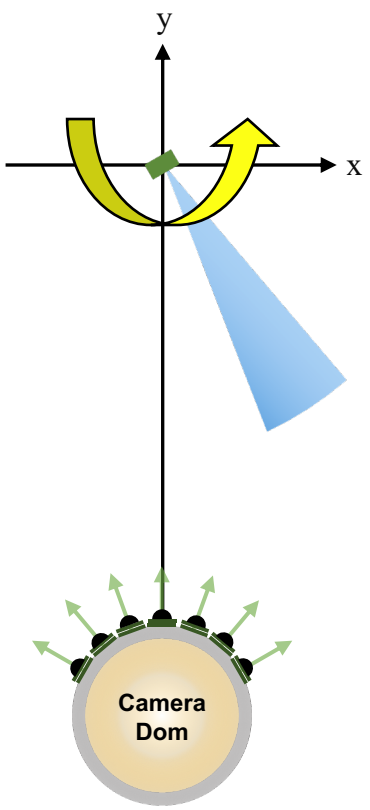


Brown et al. (2023), CNN-based Image Classification of Malicious UAVs, *Applied Sciences*.

Bulk Ice Model - *Simulation Data Set*

- In D-Eggs, the both Cameras and LEDs are spaced 120 degrees apart, so we set the camera and LED orientation into ±60 degrees, with 20-degree intervals, relative to the case where they face each other.

Geometric Setting



LED

- Position : (0, 29.8, -2200)
- Orientation : 7 cases
  > 210° ~ 330° , step = 20°
- Nphoton : 10^12
- Wave length : 470nm

Camera

- Position : (0, 0, -2200)
- Orientation : 7 cases
  > 30°~150° , step = 20°
- Field of View : 130
- Aperture : 20
- Exposure time : 10s

Total 49 Geometric Configurations

Ice Properties Setting

For each geometric configuration vary

- Bulk Ice Scattering Length : 20m ~100m (1m interval)
  - 81 cases

Total 3,969 Simulation Images

Camera Noise

For each simulation image

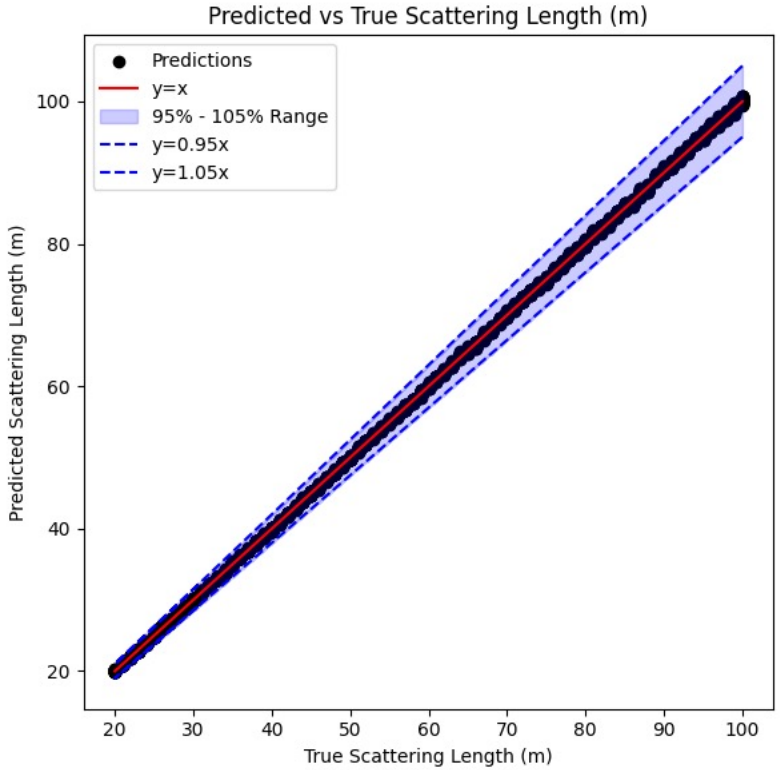
- Based on Noise Model, generate 100 noisy image

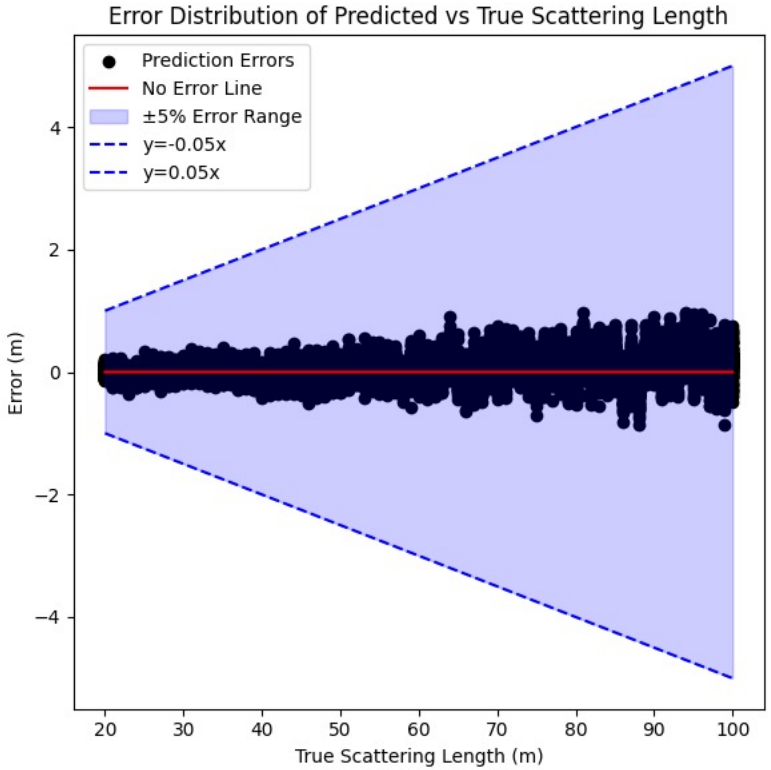
Total 396,900 Images

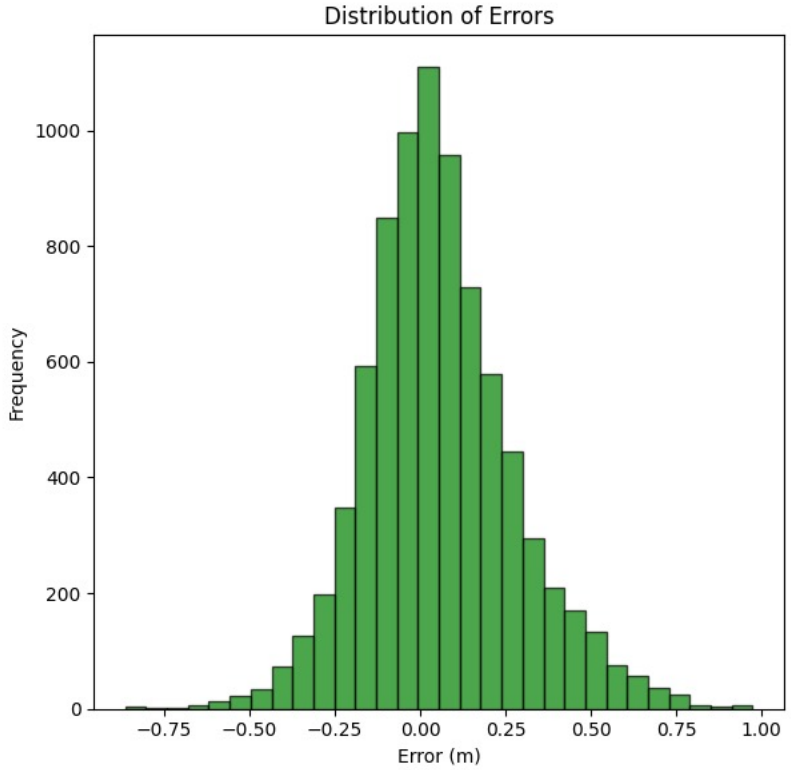
Train Image	Test Image
357,210	39,690

Bulk Ice Model – *Error Performance*

$$Error(m) = predicted\ scattering\ length(m)\ - true\ scattering\ length(m)$$



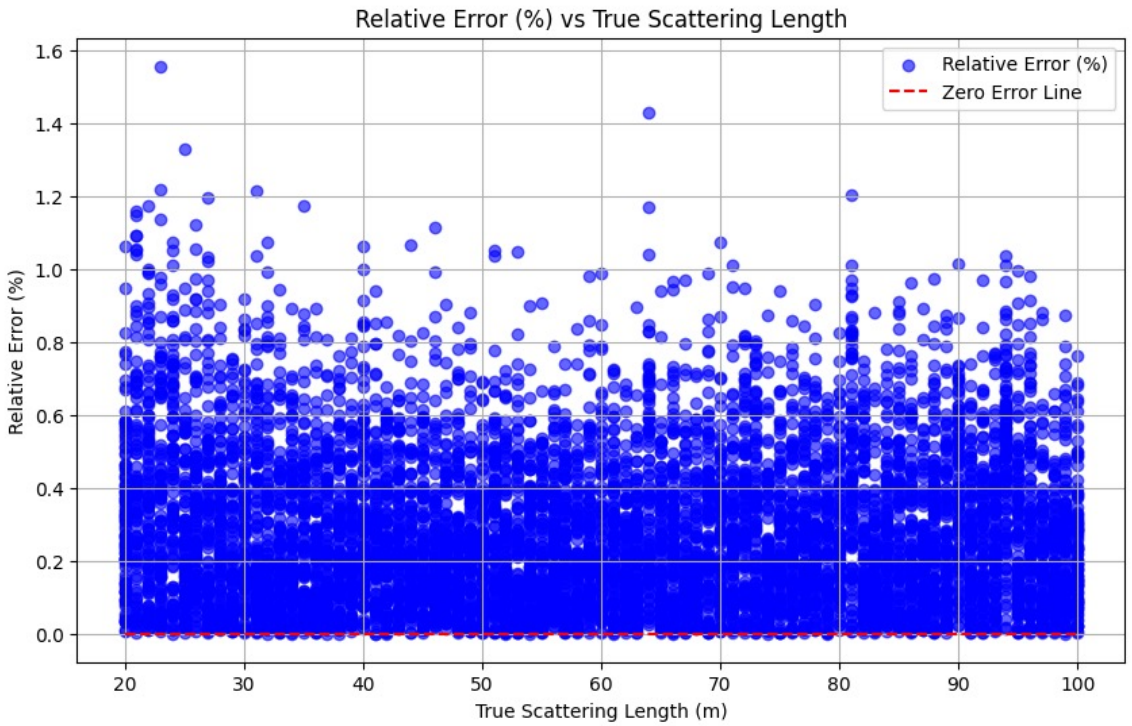




Bulk Ice Scattering Length	20m ~ 40m	40m ~ 60m	60m ~ 80m	80m ~ 100m	Total
Mean Error	0.002m	-0.003m	0.081m	0.139m	0.056m
68% Error Range	-0.159m ~ 0.200m	-0.248m ~ 0.268m	-0.282m ~ 0.377m	-0.404m ~ 0.504m	-0.273m ~ 0.337m
90% Error Range	-0.241m~0.303m	-0.400m~0.432m	-0.486m~0.650m	-0.615m~0.854m	-0.512m~0.737m

Bulk Ice Model – *Relative Error Performance*

$$Relative\ Error(\%) = \frac{|Error|}{true\ scattering\ length} * 100$$



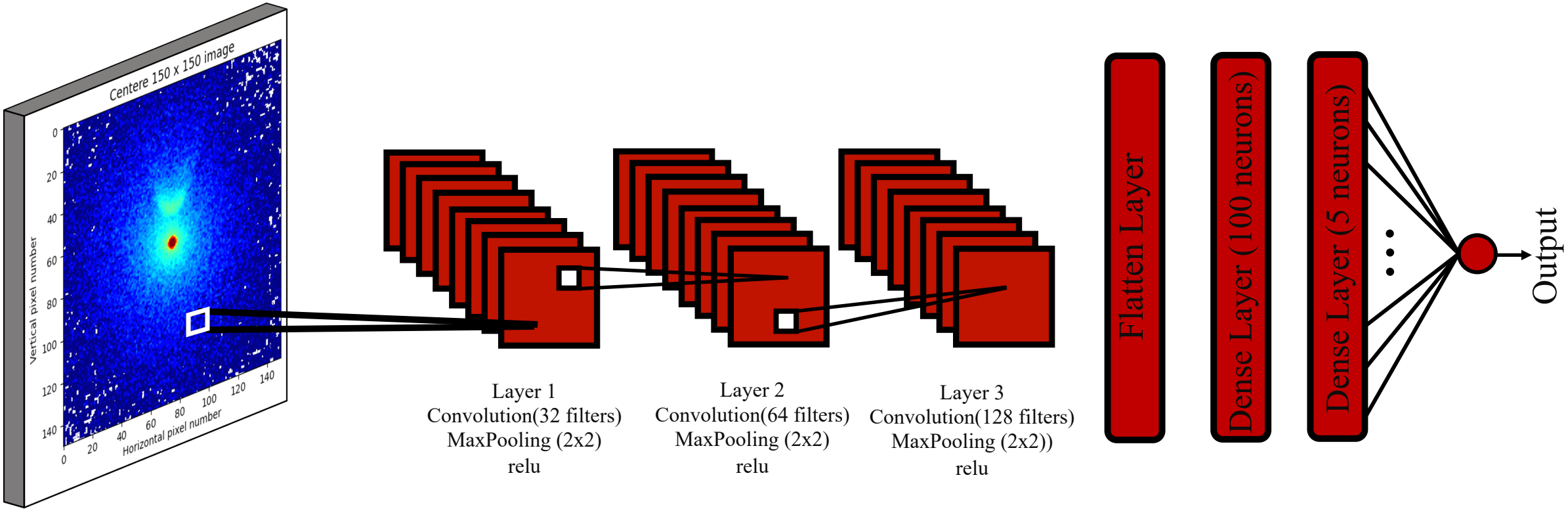
Bulk Ice Scattering Length	20m ~ 40m	40m ~ 60m	60m ~ 80m	80m ~ 100m	Total
Mean Relative Error	0.317%	0.261%	0.273%	0.295%	0.286%
68% Relative Error Range	~0.814%	~0.433%	~0.459%	~0.515%	~0.563%
90% Relative Error Range	~1.041%	~0.846%	~0.847%	~0.883%	~0.916%

Hole Ice Model – *Version 1.0*

Purpose : Measure Bubble Column Scattering length from the simulation image

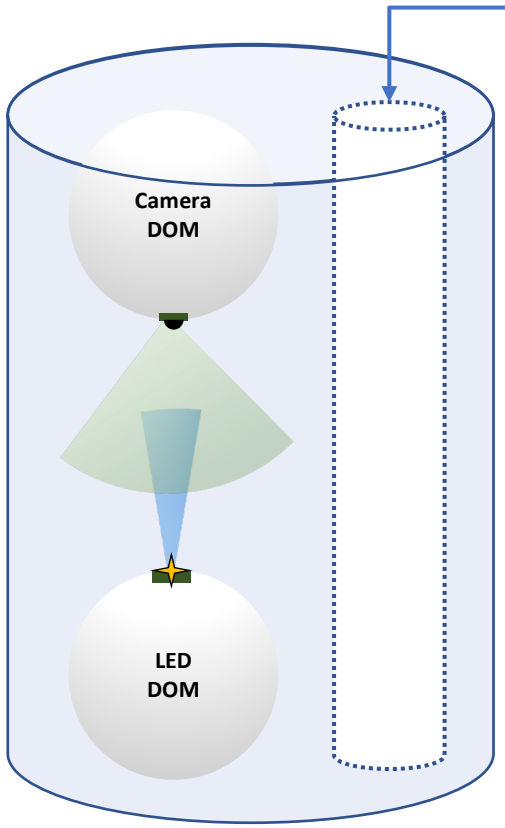
Architecture: CNN with 3 convolution layer and 2 FC layer

Main Constraint : In data setnly the bubble column scattering length is a free parameter; all other parameters, such as the bubble column offset and radius, are fixed.



Hole Ice Model - *Simulation Data Set*

Geometric Setting



Bubble Column

- Hole offset : (0, 0.2)
- Hole Radius : 0.5

unit : Dom size

Camera

- Position : (0, 0, -2200)
- Aperture : 20
- Exposure time : 3s

LED

- Position : (0, 0, -2203)
- Nphoton : 5*10^9
- Wave length : 450nm

Ice Properties Setting

- Bubble Column Scattering Length : 2cm ~ 20cm / step = 2cm
  - 10 cases
- In each case repeat Simulation 10 times

Total 100 Simulation Images

Camera Noise

For each simulation image

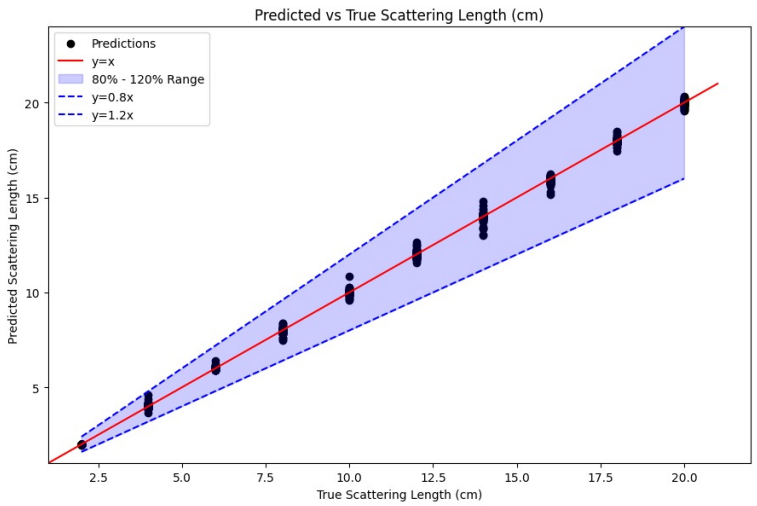
- Based on Noise Model, generate 100 noisy image

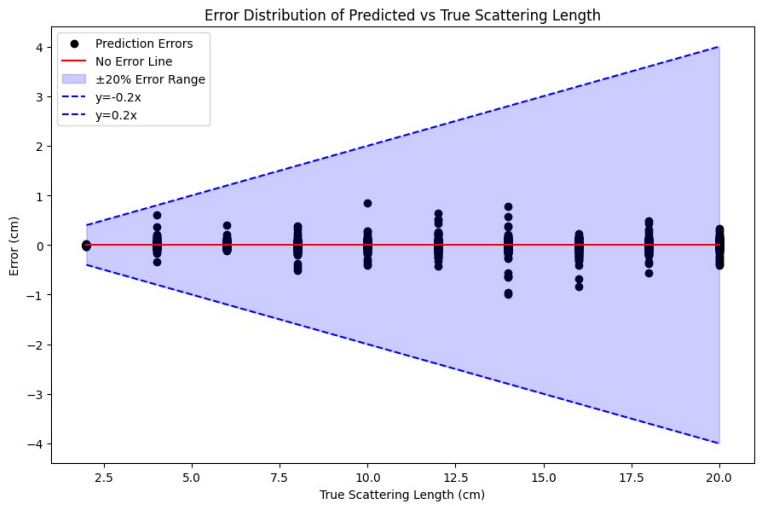
Total 10,000 Images

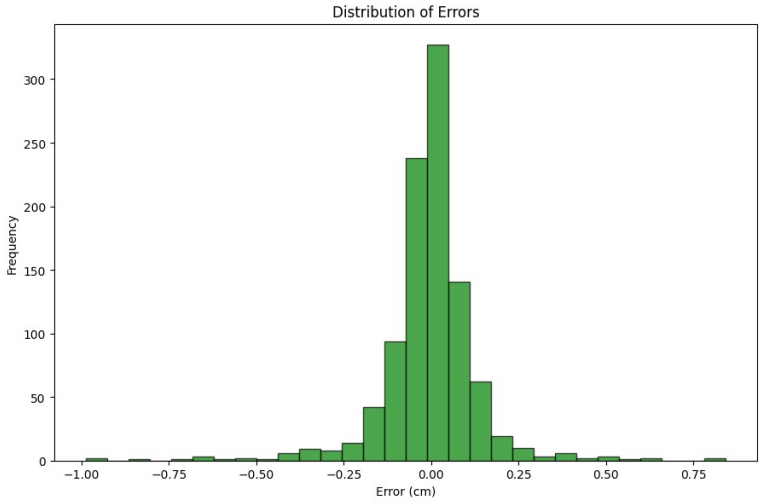
Train Image	Test Image
9000	1000

Hole Ice Model - *Performance*

$$Error(m) = predicted\ scattering\ length(m)\ - true\ scattering\ length(m)$$



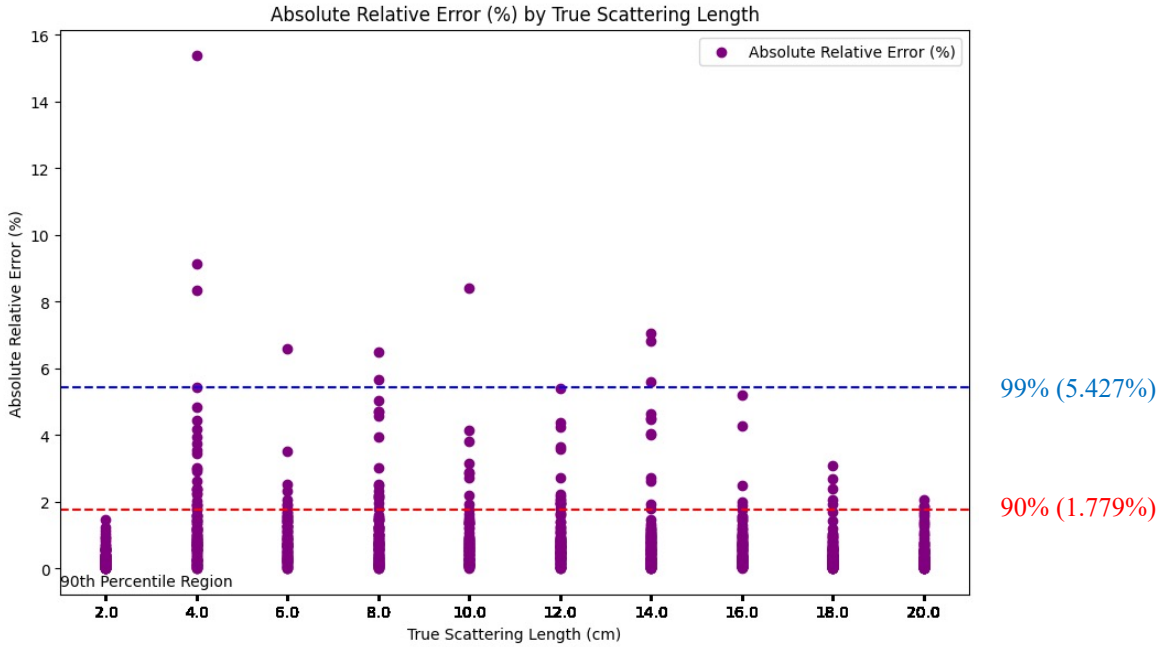




Bubble Column Scattering Length (cm)	2	4	6	8	10	12	14	16	18	20	Total
Mean of Error (cm)	-0.001	0.014	0.013	-0.018	0.003	0.009	-0.020	-0.039	-0.008	0.001	-0.004
68% Range of Error (cm)	±0.006	±0.051	±0.065	±0.081	±0.08	±0.102	±0.115	±0.118	±0.072	±0.093	±0.080
90% Range of Error (cm)	±0.013	±0.130	±0.101	±0.187	±0.183	±0.236	±0.263	±0.233	±0.169	±0.217	±0.178

Hole Ice Model - *Performance*

$$Relative\ Error(\%) = \frac{|Error|}{true\ scattering\ length} * 100$$



Bubble Column Scattering Length (cm)	2	4	6	8	10	12	14	16	18	20	Total
Mean of Relative Error (%)	0.342	1.496	0.952	1.119	0.819	0.926	0.968	0.700	0.450	0.448	0.819
68% Range of Error (%)	0.664	1.297	1.090	1.021	0.913	0.853	0.826	0.740	0.403	0.465	0.760
90% Range of Error (%)	1.226	3.268	1.683	2.348	1.835	1.972	1.882	1.458	0.941	0.1085	1.779

Summary & Conclusion

1. Total of 3,969 and 100 simulations were performed to develop the Bulk and Hole Ice models, respectively.
2. A camera noise model was applied to each simulation, generating 396,900 and 10,000 images.
3. Using these datasets, developed a ResNet-based Bulk Ice model and a CNN-based Hole Ice model.

Bulk Ice Model –1.0

	Mean	Max	90% Range
Error	0.056m	0.9739m	-0.512m~0.737m
Relative Error	0.286%	1.556%	~0.916%

Hole Ice Model –1.0

	Mean	Max	90% Range
Error	-0.004cm	0.841cm	±0.178cm
Relative Error	0.819%	15.371%	~1.779%

- The table demonstrates that each model accurately captures the baseline throughout.
- The models are expected to provide accurate measurements of bulk ice and bubble column scattering lengths

Ongoing & Future Steps

1. To upgrade the Model Bulk Ice Model, A simulate will be conducted with the horizontal difference between the Camera and LED, and the tilt of both Camera & LED.

2. To update the hole-ice model, simulations are being created with different configurations, such as the hole offset and radius.

3. The model with ViT architecture is currently under development.

Thank you for your attention

Feel free to ask any questions